



Modelling Financial Markets

Seminar 12: Options and the Volatility Surface — Hands-on



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Acronyms used in this material

- AI: Artificial Intelligence
- AR: AutoRegressive (model)
- ATM: At-The-Money
- CI: Confidence Interval
- CRR: Cox–Ross–Rubinstein (binomial tree)
- DM: Diebold–Mariano (test)
- DVOL: Deribit Volatility Index
- GMM: Generalized Method of Moments
- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- HAR-RV: Heterogeneous AutoRegressive model of Realised Volatility
- MSE: Mean Squared Error
- MZ: Mincer–Zarnowitz (regression)
- OLS: Ordinary Least Squares
- OTM: Out-of-The-Money
- QLIKE: Quasi-LIKElihood (loss function)
- SVI: Stochastic Volatility Inspired (parameterisation)
- UTC: Coordinated Universal Time (Temps Universel Coordonné)
- VIX: CBOE Volatility IndeX
- VIX3M: Cboe 3-Month Volatility Index
- VRP: Variance Risk Premium
- XFG: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)

Outline

Introduction

Part A: Derivations

Part B: Estimation and Inference

Part C: Open Question and AI Critique

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Derive no-arbitrage restrictions, the Black–Scholes equation, the Greeks and the variance-swap strike, and check them on data
- ▣ Build delta and delta-gamma hedges and measure the hedging error with a confidence interval
- ▣ Fit a smile and a risk-neutral density with bootstrap bands, and test forecasts and the variance premium with robust inference

- ▣ Background: Lecture 12; textbook [Franke, Härdle & Hafner \(2019\)](#)

Structure

- ▣ **Part A:** eight derivations, each checked on numbers (A1–A8)
- ▣ **Part B:** eight computational exercises with inference (B1–B8)
- ▣ **Part C:** an open question, a seed for the team project (C1), an AI answer to critique (C2), and a project seed from the case study of the lecture (C3)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- Volatility in annualised %; “vol points” = percentage points of volatility; time in years unless stated
- Overlapping windows are autocorrelated: always use Newey–West standard errors or a block bootstrap [Newey & West \(1987\)](#), [Künsch \(1989\)](#)

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (bounds and convexity in the strike), A3 (Black–Scholes equation and hedging P&L), A5 (one identity for the Greeks), A7 (implied volatility: existence and a starting value)
- Part B: B1 (hedging error with bootstrap intervals), B3 (SVI fit of a Bitcoin smile), B5 (variance risk premium of the S&P 500)

Proposed: you solve them

- Part A: A2 (parity as a regression; model: A1), A4 (tree convergence; model: A3), A6 (delta-gamma hedge; model: A5), A8 (variance-swap strike and its errors; model: A1)
- Part B: B2 (delta-hedged S&P 500 options; model: B1, B5), B4 (a VIX for Bitcoin; model: A8, B3), B6 (is the VIX an efficient forecast?; model: B5), B7 (Bitcoin premium; model: B5), B8 (density with bands; model: B3)
- Part C: C1, open question and project seed; C2, critique of an AI answer (Black–Scholes and parity); C3, is the option-implied bound valid and tight? (case study)

Setup: Python Environment and Data

-  **Open in Colab:** `chapter12_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `statsmodels`, `matplotlib`
- Series: S&P 500 and VIX (1990–2026), VIX3M (2007–2026); daily closes up to 18 September 2026
- Bitcoin: daily closes; DVOL (Deribit, 2021–2026); the Bitcoin option chain of the lecture (Deribit snapshot)
- Two series together: prices joined on common days first, then returns

Formulas Used Today

- Parity: $C - P = Se^{-q\tau} - Ke^{-r\tau}$; binomial: $q = (R - d)/(u - d)$, $C = [qC_u + (1 - q)C_d]/R$ Cox, Ross & Rubinstein (1979)
- Black-Scholes Black & Scholes (1973): $C = S\Phi(d_1) - Ke^{-r\tau}\Phi(d_2)$, $d_{1,2} = [\ln(S/K) + (r \pm \sigma^2/2)\tau]/(\sigma\sqrt{\tau})$
- Greeks: $\Delta = \Phi(d_1)$, $\Gamma = \varphi(d_1)/(S\sigma\sqrt{\tau})$, vega $= S\varphi(d_1)\sqrt{\tau}$, $\Theta = -S\varphi(d_1)\sigma/(2\sqrt{\tau}) - rKe^{-r\tau}\Phi(d_2)$
- Implied variance Cboe (2025): $\sigma^2 = \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{rT} Q(K_i) - \frac{1}{T} \left(\frac{F}{K_0} - 1 \right)^2$
- SVI Gatheral & Jacquier (2014): $w(k) = a + b \left(\rho(k - m) + \sqrt{(k - m)^2 + s^2} \right)$, $w = \sigma_{imp}^2 \tau$, $k = \ln(K/F)$
- Variance risk premium Carr & Wu (2009): $VRP_t = IV_t^2 - RV_{t,t+h}$, $RV_{t,t+h} = \frac{A}{h} \sum_{i=1}^h r_{t+i}^2$ (A = periods per year)

A1: Bounds and Convexity in the Strike [Solved]

□ Context and data

- ▶ European calls on a non-dividend-paying asset, price $C(K)$ for strike K , maturity τ , continuous rate r ; no arbitrage, no frictions
- ▶ Data for (d): Bitcoin calls expiring 30 October 2026 (Deribit snapshot of the lecture, 22 September 2026), BTC prices converted to USD with the Bitcoin index $S = 85\,265$ (the Deribit convention); forward $F = 85\,742$, so $e^{r\tau} = F/S$

□ Tasks

- ▶ (a) Prove $\max(S - Ke^{-r\tau}, 0) \leq C \leq S$ by comparing the pay-offs of two portfolios in every state at T
- ▶ (b) For $K_1 < K_2 < K_3$ with $K_2 = \lambda K_1 + (1 - \lambda)K_3$, prove $\lambda C(K_1) + (1 - \lambda)C(K_3) \geq C(K_2)$, i.e. C is convex in K
- ▶ (c) Show that the butterfly $[C(K - h) - 2C(K) + C(K + h)]/h^2$ tends to $e^{-r\tau}q(K)$, with q the risk-neutral density [Breedon & Litzenberger \(1978\)](#)
- ▶ (d) Compute the butterfly 84/85/86 thousand USD from mark prices, the implied density, the triangularly weighted mass $e^{r\tau}$ butterfly/ h and $P^Q(84\,000 < S_T < 86\,000)$; then count, over all 732 equally spaced call triples, how many violate convexity at mark prices and at executable prices (wings at the ask, body at the bid)

□ Report and interpret

- ▶ The bounds and the convexity proof; the butterfly value, $q(85\,000)$ and the violation counts; is any violation an arbitrage you could trade?

A1: Solution (1/2) [Solved]

- (a) Lower bound: a call plus $Ke^{-r\tau}$ in cash is worth $\max(S_T, K) \geq S_T$ at T , so it cannot cost less than the share:
 $C + Ke^{-r\tau} \geq S$
 - ▶ The call pays $(S_T - K)^+ \geq 0$, so $C \geq 0$; together $C \geq \max(S - Ke^{-r\tau}, 0)$
 - ▶ Upper bound: the share pays $S_T \geq (S_T - K)^+$ in every state, so $C \leq S$
- (b) For fixed S_T the map $K \mapsto (S_T - K)^+$ is convex, so $\lambda(S_T - K_1)^+ + (1 - \lambda)(S_T - K_3)^+ \geq (S_T - K_2)^+$ in every state
 - ▶ A portfolio that pays at least as much in every state costs at least as much today: $\lambda C(K_1) + (1 - \lambda)C(K_3) \geq C(K_2)$
- (c) $C(K) = e^{-r\tau} \int_K^\infty (x - K)q(x) dx$, so $\partial_K C = -e^{-r\tau} \int_K^\infty q(x) dx$ and $\partial_K^2 C = e^{-r\tau} q(K)$
 - ▶ The butterfly quotient is the central second difference, so it converges to $e^{-r\tau} q(K)$: convexity is exactly $q \geq 0$

A1: Solution (2/2) [Solved]

- (d) Mark prices in USD: $C(84\,000) = 4957.26$, $C(85\,000) = 4414.38$, $C(86\,000) = 3922.98$
 - ▶ Butterfly = $4957.26 - 2 \cdot 4414.38 + 3922.98 = 51.47$ USD, positive as required
 - ▶ $e^{r\tau} = F/S = 1.0056$: $q(85\,000) \approx 1.0056 \cdot 51.47 / 1\,000^2 = 5.18 \cdot 10^{-5}$ per USD
 - ▶ The butterfly pays $(h - |S_T - K|)^+$: $e^{r\tau}$ butterfly/ $h = 5.2\%$ is a triangularly weighted mass $\approx h q(K)$; $P^Q(84\,000 < S_T < 86\,000) \approx 2h q(K) = 10.4\%$
- All triples: 7 of 732 mark butterflies are negative, all among deep in-the-money calls, the worst -11.80 USD
 - ▶ The median bid–ask spread at those strikes is 255.80 USD; at executable prices the number of negative butterflies is 0
- Interpretation: no tradable arbitrage; but a density estimated from raw marks would be negative in places, which is why the lecture smooths the smile (SVI) before differentiating

A2: Parity as a Regression: the Implied Forward [Proposed]

Solution discussed in class

□ Context and data

- ▶ European calls and puts with the same expiry and strikes $K_1, \dots, K_n$; discount factor $D = e^{-r\tau}$ and forward F unknown
- ▶ Data: the expiry of A1 (30 October 2026), USD mid prices $(bid + ask)/2 \times S$ (S the Bitcoin index, the Deribit convention), only strikes where both the call and the put have a bid and an ask

□ Tasks

- ▶ (a) Show from put–call parity that $C(K) - P(K) = DF - DK$ for every strike
- ▶ (b) Write the OLS regression $C_i - P_i = a + bK_i + e_i$; derive $\hat{D}$, $\hat{F}$ from $(\hat{a}, \hat{b})$ and the delta-method standard error of $\hat{F}$ with heteroskedasticity-robust covariance
- ▶ (c) Estimate on the data; compare $\hat{F}$ with the exchange forward in standard errors

□ Report and interpret

- ▶ $\hat{D}$, $\hat{F}$ with standard errors, the number of strikes; which USD interest rate does $\hat{D}$ imply, and why might $\hat{F}$ differ from the exchange forward?
- ▶ Model: A1 (portfolios with identical pay-offs)

A3: The Black–Scholes Equation and the Hedging P&L [Solved]

□ Context and data

- ▶ Price $dS_t = \mu S_t dt + \sigma S_t dW_t$, constant r , continuous trading without costs; option value $V(S, t)$, twice differentiable [Black & Scholes \(1973\)](#)
- ▶ Numbers for (c): the call of A5 ($S = K = 100$, six months, $r = 3\%$), sold at $\sigma_{imp} = 25\%$, gamma $\Gamma = 0.0222$; 252 trading days a year

□ Tasks

- ▶ (a) Replicate V with a self-financing portfolio of Δ shares and cash B at r ($d\Pi = \Delta dS + rB dt$); apply Itô's lemma to V , choose Δ to remove the dW_t term and use no arbitrage to derive the Black–Scholes equation
- ▶ (b) A dealer sells the option at σ_{imp} , holds $\Delta = \partial_S V^{imp}$ shares and cash at r , while S moves with σ_{real} ; derive the instantaneous P&L
- ▶ (c) Compute today's P&L rate per trading day if $\sigma_{real} = 30\%$, then annualise it with S and Γ frozen at today's values

□ Report and interpret

- ▶ The equation, the P&L formula and the two numbers; who gains and why does the P&L depend on the path of S ?

A3: Solution (1/2) [Solved]

- (a) Itô: $dV = (V_t + \mu SV_S + \frac{1}{2}\sigma^2 S^2 V_{SS}) dt + \sigma SV_S dW_t$; the self-financing $\Pi = \Delta S + B$, started at $\Pi = V$, has $d\Pi = \Delta dS + rB dt$
- $d(V - \Pi) = (V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} - rB) dt + (V_S - \Delta)(\mu S dt + \sigma S dW_t)$; with $\Delta = V_S$ both the risk and μ disappear
- A riskless position that costs 0 earns 0: $V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} = rB = r(V - SV_S)$, i.e. $V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0$
- (b) The self-financing book $-V + \Delta S + (V - \Delta S)$ in cash changes by $-dV + \Delta dS + r(V - \Delta S) dt$, plus interest on the P&L already accumulated
- Itô with the true volatility: $dV = V_t dt + \Delta dS + \frac{1}{2}\Gamma \sigma_{real}^2 S^2 dt$; the equation at σ_{imp} gives $V_t = rV - rS\Delta - \frac{1}{2}\sigma_{imp}^2 S^2 \Gamma$

A3: Solution (2/2) [Solved]

- Substituting, the dS and r terms cancel: $d \text{P\&L} = r \text{P\&L} dt + \frac{1}{2} \Gamma S^2 (\sigma_{imp}^2 - \sigma_{real}^2) dt$; the second term is the excess gain
- (c) $\frac{1}{2} \cdot 0.02223 \cdot 100^2 \cdot (0.25^2 - 0.30^2) \cdot \frac{1}{252} = \frac{1}{2} \cdot 0.02223 \cdot 10^4 \cdot (-0.0275)/252 = -0.0121$ per option and trading day
- Annualised with S and Γ frozen: -3.06 , about 39% of the call price 7.76; a rate, not the realised P&L of this six-month option, since ΓS^2 changes along the path
- Interpretation: the seller loses when realised volatility exceeds implied; the loss accrues through Γ , largest near the strike and near expiry, so two paths with the same realised variance give different P&L

A4: The Binomial Tree Converges to Black–Scholes [Proposed]

Solution discussed in class

□ Context and data

- ▶ Cox–Ross–Rubinstein (CRR) tree with N steps of length $\Delta t = T/N$ Cox, Ross & Rubinstein (1979); European call of the lecture: $S = K = 100$, $T = 1$, $r = 5\%$, $\sigma = 20\%$

□ Tasks

- ▶ (a) Show that $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$, $q = (e^{r\Delta t} - d)/(u - d)$ match the risk-neutral mean exactly and the variance of $\ln(S_{t+\Delta t}/S_t)$ up to terms of order Δt^2
- ▶ (b) Price the call with $N = 50, 51, 100, 101, 200, 201$ steps and compute $N(C_N - C_{BS})$ for each
- ▶ (c) Average the prices for $N = 200$ and $N = 201$ and compare the error with that of $N = 200$

□ Report and interpret

- ▶ A table of C_N and $N(C_N - C_{BS})$; the order of the error, the reason for the even/odd oscillation, and why averaging helps
- ▶ Model: A3 (from the tree to the continuous-time equation)

A5: One Identity for All the Greeks [Solved]

□ Context and data

- ▶ Black–Scholes call $C = S\Phi(d_1) - Ke^{-r\tau}\Phi(d_2)$, $d_{1,2} = [\ln(S/K) + (r \pm \sigma^2/2)\tau]/(\sigma\sqrt{\tau})$; φ = standard Normal density
- ▶ Numbers for (c): $S = K = 100$, six months ($\tau = 0.5$), $r = 3\%$, $\sigma = 25\%$, no dividends

□ Tasks

- ▶ (a) Prove $S\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$
- ▶ (b) Use (a) to derive $\Delta = \partial_S C = \Phi(d_1)$ and vega $= \partial_\sigma C = S\varphi(d_1)\sqrt{\tau}$; then show vega $= \Gamma S^2\sigma\tau$ with $\Gamma = \varphi(d_1)/(S\sigma\sqrt{\tau})$
- ▶ (c) Compute d_1 , d_2 , both sides of (a), Δ , Γ , vega per unit of σ in two ways, the call and the put (from parity)

□ Report and interpret

- ▶ The proofs and the numerical checks; what does vega $= \Gamma S^2\sigma\tau$ imply for hedging gamma with options of the same expiry?

A5: Solution (1/2) [Solved]

- (a) $\varphi(d_2)/\varphi(d_1) = e^{(d_1^2 - d_2^2)/2}$ and $d_1^2 - d_2^2 = (d_1 - d_2)(d_1 + d_2) = \sigma\sqrt{\tau} \cdot \frac{2 \ln(S/K) + 2r\tau}{\sigma\sqrt{\tau}}$
- Hence $\varphi(d_2)/\varphi(d_1) = Se^{r\tau}/K$, i.e. $S\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$
- (b) $\partial_S C = \Phi(d_1) + [S\varphi(d_1) - Ke^{-r\tau}\varphi(d_2)] \partial_S d_1 = \Phi(d_1)$, since $\partial_S d_1 = \partial_S d_2$ and the bracket is 0 by (a)
- Vega: $S\varphi(d_1)\partial_\sigma d_1 - Ke^{-r\tau}\varphi(d_2)\partial_\sigma d_2 = S\varphi(d_1)\partial_\sigma(d_1 - d_2) = S\varphi(d_1)\sqrt{\tau}$
- And $\Gamma S^2 \sigma \tau = \frac{\varphi(d_1)}{S\sigma\sqrt{\tau}} S^2 \sigma \tau = S\varphi(d_1)\sqrt{\tau} = \text{vega}$

A5: Solution (2/2) [Solved]

- (c) $\sigma\sqrt{\tau} = 0.1768$; $d_1 = (0 + (0.03 + 0.03125) \cdot 0.5)/0.1768 = 0.1732$, $d_2 = -0.0035$
- Identity: $S\varphi(d_1) = 100 \cdot 0.3930 = 39.30$ and $Ke^{-r\tau}\varphi(d_2) = 98.51 \cdot 0.3989 = 39.30$: equal
- $\Delta = \Phi(0.1732) = 0.5688$; $\Gamma = 0.3930/(100 \cdot 0.1768) = 0.0222$
- $\text{Vega} = 100 \cdot 0.3930 \cdot \sqrt{0.5} = 27.79$ and $\Gamma S^2 \sigma \tau = 0.02223 \cdot 10^4 \cdot 0.25 \cdot 0.5 = 27.79$ per unit of σ (0.2779 per vol point)
- $C = 100 \cdot 0.5688 - 98.51 \cdot 0.4986 = 7.76$; parity: $P = C - 100 + 98.51 = 6.27$
- Interpretation: for one expiry, vega and gamma are proportional, so a gamma-neutral book of same-expiry options is also vega-neutral, but only while the smile is flat (A6)

A6: A Delta-Gamma Hedge [Proposed]

Solution discussed in class

□ Context and data

- ▶ A dealer has **sold** 1 000 calls of A5 ($S = K = 100$, six months, $r = 3\%$, $\sigma = 25\%$)
- ▶ Hedging instruments: a call with $K = 110$, same expiry, Black–Scholes price 3.90, and the stock; Greeks from the Black–Scholes formulas

□ Tasks

- ▶ (a) Compute the delta and the gamma of the dealer's position
- ▶ (b) Find the number n_2 of $K = 110$ calls that makes the gamma zero, then the number of shares that makes the delta zero
- ▶ (c) Compute the vega of the hedged book and explain the result

□ Report and interpret

- ▶ The positions in the two instruments and the residual vega; is the book protected against a change in volatility?
- ▶ Model: A5

A7: Implied Volatility: Existence, Uniqueness, Starting Value [Solved]

□ Context and data

- ▶ European call, Black–Scholes price $BS(\sigma)$; market price 4.50 for $S = K = 100$, three months ($\tau = 0.25$), $r = 2\%$

□ Tasks

- ▶ (a) Prove that $\sigma \mapsto BS(\sigma)$ is continuous and strictly increasing, with limits $(S - Ke^{-r\tau})^+$ as $\sigma \rightarrow 0$ and S as $\sigma \rightarrow \infty$; conclude when an implied volatility exists
- ▶ (b) Derive the at-the-money-forward approximation $C \approx S\sigma\sqrt{\tau}/\sqrt{2\pi}$ [Brenner & Subrahmanyam \(1988\)](#)
- ▶ (c) Use (b), with and without subtracting half the discounted forward intrinsic value $(S - Ke^{-r\tau})/2$, as a starting value; then run one Newton step $\sigma_1 = \sigma_0 - (BS(\sigma_0) - 4.50)/\mathcal{V}(\sigma_0)$ from $\sigma_0 = 20\%$

□ Report and interpret

- ▶ The two starting values, σ_1 and the exact implied volatility; why does Newton converge so fast here, and when would it fail?

A7: Solution (1/2) [Solved]

- (a) $\partial_{\sigma} BS = S\varphi(d_1)\sqrt{\tau} > 0$ (A5): strictly increasing and continuous
- $\sigma \rightarrow 0$: $d_{1,2} \rightarrow \pm\infty$ according to the sign of $\ln(S/K) + r\tau$, so $BS \rightarrow (S - Ke^{-r\tau})^+$; $\sigma \rightarrow \infty$: $d_1 \rightarrow \infty$, $d_2 \rightarrow -\infty$, so $BS \rightarrow S$
- By the intermediate value theorem a unique σ_{imp} exists iff the price lies strictly inside the no-arbitrage bounds of A1
- (b) If $S = Ke^{-r\tau}$: $d_{1,2} = \pm\sigma\sqrt{\tau}/2$ and $C = S[\Phi(\sigma\sqrt{\tau}/2) - \Phi(-\sigma\sqrt{\tau}/2)] \approx S\sigma\sqrt{\tau}\varphi(0) = S\sigma\sqrt{\tau}/\sqrt{2\pi}$

A7: Solution (2/2) [Solved]

- (c) Without correction: $\sigma \approx 4.50\sqrt{2\pi}/(100 \cdot 0.5) = 22.56\%$
- With correction: $Ke^{-rT} = 99.501$, half the discounted forward intrinsic value 0.249, $\sigma \approx (4.50 - 0.249)\sqrt{2\pi}/50 = 21.31\%$
- Newton: $BS(20\%) = 4.2322$, $\mathcal{V} = 19.848$, so $\sigma_1 = 0.20 - (4.2322 - 4.50)/19.848 = 21.3495\%$; exact 21.3495%
- Interpretation: near the money the price is almost linear in σ , so one Newton step gives four decimals, and the corrected approximation is already within 0.05 vol points
- Deep out of the money vega is close to 0 and Newton can overshoot; there, use a bracketing solver (Brent) inside the bounds

A8: The Variance-Swap Strike and Its Estimation Error [Proposed]

Solution discussed in class

□ Context and data

- ▶ Spanning formula of the lecture: $f(S_T) = f(F) + f'(F)(S_T - F) + \int_0^F f''(K)(K - S_T)^+ dK + \int_F^\infty f''(K)(S_T - K)^+ dK$
Bakshi, Kapadia & Madan (2003)
- ▶ Log-normal benchmark: $F = 100$, $r = 0$, 90 days ($T = 90/365$), flat volatility 20%, so the exact variance is 0.04; OTM prices from Black–Scholes

□ Tasks

- ▶ (a) Apply the formula with $f(S) = -2 \ln(S/F)$ and show $K_{var} = \frac{2e^{rT}}{T} \left[\int_0^F \frac{P(K)}{K^2} dK + \int_F^\infty \frac{C(K)}{K^2} dK \right]$ for continuous paths
- ▶ (b) Compute the Cboe sum with $\Delta K = 5$ on $[80, 120]$ Cboe (2025), a dense integral on $[80, 120]$ and a dense integral on $(0, \infty)$
- ▶ (c) Split the total error into discretisation (Cboe minus dense on $[80, 120]$) and truncation (dense on $[80, 120]$ minus exact)

□ Report and interpret

- ▶ The three variances and volatilities and the two errors; which dominates here, and what changes for a longer maturity?
- ▶ Model: A1 (portfolios of options as prices of pay-off shapes)

B1: Hedging Error and Rebalancing Frequency [Solved]

▣ Context and data

- ▶ Simulated data: 10 000 paths of geometric Brownian motion, $S_0 = 100$, $\mu = 8\%$, $\sigma = 20\%$, 252 steps over three months (seed 42)
- ▶ A three-month ATM call ($K = 100$, $r = 4\%$) is sold at its Black–Scholes price and delta-hedged at $\sigma = 20\%$

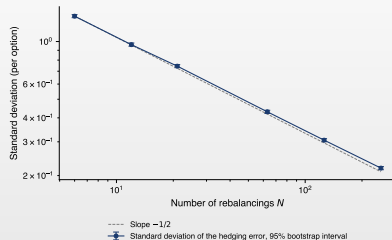
▣ Tasks

- ▶ (a) Rebalance $N = 6, 12, 21, 63, 126, 252$ times; compute the standard deviation of the hedging error at expiry for each N
- ▶ (b) Build 95% percentile bootstrap intervals for each standard deviation from 1 000 resamples of the paths [Efron \(1979\)](#)
- ▶ (c) Regress $\ln(\text{s.d.})$ on $\ln N$; give the slope with a bootstrap interval (the same 1 000 resamples)

▣ Report and interpret

- ▶ A table of s.d. and intervals and a log-log chart; is the slope consistent with $-1/2$? How often must a dealer rebalance to keep the standard deviation of the error below 10% of the premium? What would transaction costs change?

B1: Solution (1/2) [Solved]



N	6	12	21	63	126	252
s.d.	1.356	0.963	0.744	0.430	0.306	0.219
95% CI	[1.331, 1.381]	[0.945, 0.983]	[0.731, 0.758]	[0.422, 0.438]	[0.300, 0.312]	[0.215, 0.223]

B1: Solution (2/2) [Solved]

- Step 1: for each path, sell the call at the premium 4.49, buy Δ_0 shares, finance with cash at r ; at each rebalancing date reset Δ ; at expiry, error = shares + cash - pay-off
- Step 2: the s.d. falls from 1.356 ($N = 6$) to 0.219 ($N = 252$); the intervals are narrow because 10 000 paths give a precise s.d.
- Step 3: slope -0.489 , 95% CI $[-0.496, -0.482]$: close to $-1/2$, slightly above it at these finite N (the $1/\sqrt{N}$ law is asymptotic)
- Reading: daily hedging leaves 9.6% of the premium, about the 10% target; four times a day halves it again (ratio 1.97)
- With a fixed fee per trade, costs grow like N while the error falls like $1/\sqrt{N}$; the optimal frequency depends on the cost model and on the chosen trade-off between risk and cost

B2: Delta-Hedged S&P 500 Options, 1990–2026 [Proposed]

Solution discussed in class

□ Context and data

- ▶ Daily S&P 500 closes and VIX (GSPC.INDX, VIX.INDX in data/market), January 1990 to September 2026; interest and dividends ignored
- ▶ Every 21 trading days, sell a one-month ATM call priced with the VIX and hedge it daily at the same volatility (the stylised back-test of the lecture: VIX, not ATM quotes) [Bakshi & Kapadia \(2003\)](#); P&L in % of the index

□ Tasks

- ▶ (a) Compute the mean monthly P&L and its Newey–West standard error with 3 lags; test H_0 : mean = 0 at the 5% level
- ▶ (b) Build a 95% moving-block bootstrap interval for the mean (blocks of 6 months, 2 000 resamples) [Künsch \(1989\)](#)
- ▶ (c) Compute the share of positive months, the skewness and the share of the total gain lost in the five worst months

□ Report and interpret

- ▶ The mean with both intervals and the tail statistics; is the premium significant? Why can a strategy with a high t -statistic still be dangerous? What would interest and dividends change?
- ▶ Model: B1 (bootstrap), B5 (Newey–West)

B3: An SVI Smile for Bitcoin [Solved]

□ Context and data

- ▶ Deribit snapshot of the lecture (ch12_deribit_snapshot.csv); expiry closest to 30 days: 2026-10-30, 38 days, 56 quoted OTM options (puts below the forward, calls above)
- ▶ Total variance $w(k) = \sigma_{\text{mark}}^2 \tau$ at log-moneyness $k = \ln(K/F)$; SVI $w(k) = a + b(\rho(k - m) + \sqrt{(k - m)^2 + s^2})$ Gatheral & Jacquier (2014)

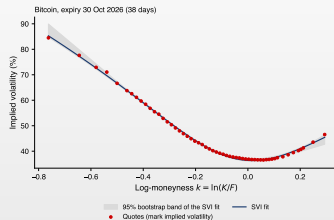
□ Tasks

- ▶ (a) Fit the five SVI parameters by least squares on w , with $b \geq 0$, $|\rho| < 1$, $s > 0$; report the fit error in vol points
- ▶ (b) Pairs bootstrap (300 resamples of the quotes, refit each time): 95% intervals for the ATM volatility and the skew $IV(0.15) - IV(-0.15)$
- ▶ (c) Evaluate the butterfly function $g(k)$ of Gatheral & Jacquier (2014) on $k \in [-1.5, 1.5]$ and report its minimum

□ Report and interpret

- ▶ Parameters, intervals and $\min g$; which parameter controls the skew and what is its sign? Is the skew significantly different from 0? Why fit w and not the prices?

B3: Solution (1/2) [Solved]



- (a) $\hat{a} = -0.020$, $\hat{b} = 0.102$, $\hat{\rho} = -0.262$, $\hat{m} = -0.079$, $\hat{s} = 0.348$; root mean squared error 0.42 vol points

B3: Solution (2/2) [Solved]

- (b) ATM volatility 36.37%, 95% CI [36.10, 36.87]; skew -2.11 , 95% CI $[-2.73, -1.52]$
- ρ controls the tilt: $\hat{\rho} < 0$ makes the put wing dearer; the skew interval excludes 0, so the tilt is significant at 5%
- (c) $\min_k g(k) = 0.136 > 0$ on $k \in [-1.5, 1.5]$: no butterfly arbitrage is detected on this grid; a positive density everywhere also needs $g \geq 0$ outside the grid and the wing conditions of [Gatheral & Jacquier \(2014\)](#)
- Why w : it is comparable across strikes and nearly linear in the wings, while option prices span four orders of magnitude and would weight the fit towards the dearest options
- Caveat: the bootstrap resamples smoothed mark volatilities, so the intervals measure fitting noise, not bid–ask uncertainty

B4: A VIX for Bitcoin [Proposed]

Solution discussed in class

□ Context and data

- ▶ Same snapshot; the two expiries around 30 days (2026-10-09, 2026-10-30); USD mid price $Q = \text{BTC mid} \times \text{index } S$ (the Deribit convention); F and $e^{RT} = F/S$ from the Deribit forward

□ Tasks

- ▶ (a) Compute the implied variance of each expiry with the Cboe formula [Cboe \(2025\)](#): K_0 = first strike at or below F ; puts below K_0 , calls above, the put–call average at K_0 ; no zero bids, each wing stops after two consecutive zero bids; ΔK_i = half the distance between neighbouring strikes, one-sided at the ends
- ▶ (b) Interpolate the total variances linearly in time to 30 days and take $100\sigma_{30}$
- ▶ (c) Compare with the ATM volatilities of the two expiries and with DVOL at the same moment [Alexander & Imeraj \(2020\)](#)

□ Report and interpret

- ▶ The two variances, the weight of the near expiry and the index; why is the index above both ATM volatilities, and which choices (strike cut-off, mid vs mark prices) could explain the gap to DVOL?
- ▶ Model: A8 (strip formula and its errors), B3 (the same snapshot)

B5: The Variance Risk Premium of the S&P 500 [Solved]

□ Context and data

- ▶ Daily S&P 500 and VIX closes, joined on common days, January 1990 to August 2026 (last window ending in September 2026)
- ▶ Ex-post premium $VRP_t = VIX_t^2 - RV_{t,t+21}$, $RV_{t,t+21} = \frac{252}{21} \sum_{i=1}^{21} r_{t+i}^2$ in %² Carr & Wu (2009); vol points: $VIX_t - \sqrt{RV_{t,t+21}}$

□ Tasks

- ▶ (a) Compute the mean VRP, its Newey–West standard error with 21 lags Newey & West (1987), a 95% interval and the naive standard error $s/\sqrt{n}$
- ▶ (b) Repeat in vol points and compute the share of days with a positive premium
- ▶ (c) Repeat for 1990–2007 and 2008–2026, test each subperiod mean at the 5% level, then test their equality: regress VRP_t on a constant and a 2008–2026 dummy (Newey–West, 21 lags)

□ Report and interpret

- ▶ A table by period; why is the naive standard error wrong here, and by how much? Is the premium stable across subperiods? Why is the variance premium less precise than the vol-point premium?

B5: Solution (1/2) [Solved]

	n	Mean VRP (% ²)	95% CI	Vol points (s.e.)	Positive
1990–2026	9 203	113.1	[72.7, 153.6]	4.09 (0.27)	86%
1990–2007	4 516	149.3	[125.2, 173.5]	4.60 (0.25)	88%
2008–2026	4 687	78.3	[2.7, 153.9]	3.60 (0.46)	84%

- Step 1: build $RV_{t,t+21}$ from the next 21 log returns, then VRP_t on every day with a complete window
- Step 2: regress VRP_t on a constant with HAC (Bartlett, 21 lags: the overlap; check longer bandwidths) standard errors; interval = mean ± 1.96 s.e.

B5: Solution (2/2) [Solved]

- (a) Full sample: $t = 113.1/20.6 = 5.49$: reject H_0 at 5%; naive s.e. 5.3 vs Newey–West 20.6: 3.9 times too small, because consecutive windows share 20 of 21 days
- (b) In vol points the premium is 4.09 (s.e. 0.27), positive on 86% of days
- (c) 1990–2007: $t = 12.12$; 2008–2026: $t = 2.03$: both reject at 5%, but after 2008 barely, because of the crisis windows of 2008 and 2020
- Equality: difference $-71.0\%^2$ (s.e. 40.5, $t = -1.76$, $p = 0.08$); in vol points -1.01 ($t = -1.92$, $p = 0.06$)
- Reading: squaring magnifies the crisis windows, so the variance premium has heavier tails and a wider interval than the vol-point premium; the premium is positive in both subperiods, but the difference in size is not significant at 5%: the data cannot tell whether it changed

B6: Is the VIX an Efficient Forecast? [Proposed]

Solution discussed in class

Context and data

- ▶ The daily series of B5: $RV_{t,t+21}$ and VIX_t^2 , 1990–2026; for (d) also daily squared returns (annualised, in %²) for a log HAR-RV: $\ln RV_{t,t+21}$ on the logs of their 1, 5 and 22-day averages, each floored at 0.01 [Corsi \(2009\)](#), level forecast $\exp(\widehat{\ln RV} + \hat{s}^2/2)$, $\hat{s}^2$ the residual variance

Tasks

- ▶ (a) Estimate $RV_{t,t+21} = \alpha + \beta VIX_t^2 + \varepsilon_t$ [Mincer & Zarnowitz \(1969\)](#) with Newey–West errors (21 lags); test $\alpha = 0$ and $\beta = 1$ separately and jointly (Wald, 5%); report $\text{corr}(\hat{\alpha}, \hat{\beta})$
- ▶ (b) Add $RV_{t-21,t}$; then re-estimate the regression in logs, $\ln RV$ on $\ln VIX^2$, and test $\beta = 1$
- ▶ (c) Instrument $\ln VIX_t^2$ with $\ln VIX_{t-21}^2$ [Christensen & Prabhala \(1998\)](#) (just-identified GMM, HAC covariance); report the first-stage t and test $\beta = 1$
- ▶ (d) From 2000, re-estimate every 21 days on an expanding window of targets already observed; compare VIX^2 with HAR-RV by QLIKE loss and a Diebold–Mariano test with HAC (21 lags) [Diebold & Mariano \(1995\)](#), [Patton \(2011\)](#)

Report and interpret

- ▶ All estimates and tests with decisions at 5%; why can both t -statistics be small while the joint test rejects? How do the conclusions change in logs and with instruments, and what must the instrument satisfy? Does in-sample significance survive out of sample?
- ▶ Model: B5 (Newey–West inference)

B7: The Variance Risk Premium of Bitcoin [Proposed]

Solution discussed in class

□ Context and data

- ▶ DVOL (Deribit), March 2021 to August 2026, and daily Bitcoin closes (BTC-USD.CC); realised volatility over the next 30 calendar days, $A = 365$

□ Tasks

- ▶ (a) Compute the mean premium $DVOL_t - \sqrt{RV_{t,t+30}}$ in vol points, its Newey–West standard error with 30 lags, and test it at 5%
- ▶ (b) Compute the share of positive days and the ratio mean implied / mean realised volatility; compare both with the S&P 500 of B5
- ▶ (c) Count the non-overlapping 30-day periods in the sample

□ Report and interpret

- ▶ Premium, test, ratios and effective sample size; is the Bitcoin premium larger or smaller than the equity premium, and in which unit? How much can five years of data tell us?
- ▶ Model: B5

B8: A Risk-Neutral Density with Confidence Bands [Proposed]

Solution discussed in class

□ Context and data

- ▶ Deribit snapshot of the lecture; expiry closest to 90 days: 2026-12-25, 94 days, 58 OTM quotes; SVI fitted as in B3; forward (undiscounted) prices

□ Tasks

- ▶ (a) Compute the density $q(K) = \partial^2 C / \partial K^2$ from the SVI call prices (Black-76 with F) on a fine strike grid [Breedon & Litzenberger \(1978\)](#)
- ▶ (b) Compute $P^Q(S_T < 0.8F)$ and $P^Q(S_T > 1.2F)$ with 95% pairs-bootstrap intervals (300 refits) and the log-normal values at the ATM volatility; in each refit recompute the log-normal value at the ATM volatility of that refit and give a 95% interval for the difference SVI minus log-normal
- ▶ (c) Report where the quotes end in K/F and which part of each probability relies on extrapolated wings

□ Report and interpret

- ▶ The two probabilities with intervals vs log-normal; is the crash probability significantly above or below the log-normal one? What does the bootstrap band not measure?
- ▶ Model: B3 (SVI fit and pairs bootstrap)

C1: Is the Variance Risk Premium a Tradable Signal?

Question

- Would you sell variance on the S&P 500 or on Bitcoin every month, and when would you stop? Carr & Wu (2009), Bollerslev, Tauchen & Zhou (2009)
- Instrument: a synthetic variance swap with vega notional 1, $P\&L = (IV^2 - RV)/(2 IV)$ vol points per month
- S&P 500 with the VIX (1990–2026); Bitcoin with DVOL (2021–2026)

What a good answer contains

- Sharpe ratio with a block-bootstrap interval, skewness, worst month, maximum drawdown
- A timing rule, e.g. skip months that start in backwardation (VIX above VIX3M); is it better out of sample?
- Comparison with the delta-hedged option of the lecture; position sizing and margin
- The ex-ante VRP ($VIX_t^2/12 - RV_t$) as a predictor of 1–12-month excess returns, with Hodrick (1992) errors and a bootstrap under the null Bollerslev, Tauchen & Zhou (2009), Hodrick (1992)

C2: Critique an AI Answer [Proposed]

- A student asks an AI assistant: “Price a European call with 90 calendar days to expiry on a non-dividend-paying stock with $S = 100$, $K = 105$, volatility 20%, continuous rate $r = 4\%$; then give the put from put–call parity”
- The assistant answers with this code:

```
import numpy as np
from scipy.stats import norm
S, K, r, sigma = 100.0, 105.0, 0.04, 20          # sigma = 20%
T = 90 / 252                                     # 90 days
d1 = (np.log(S / K) + (r + 0.5 * sigma**2) * T) / (sigma * np.sqrt(T))
d2 = d1 - sigma * np.sqrt(T)
call = S * norm.cdf(d1) - K * np.exp(-r * T) * norm.cdf(d2)
put = call + S - K * np.exp(-r * T)             # put-call parity
print(f"call_{S}={call:.2f},_{K}put_{S}={put:.2f}")
```

- “The code follows Black–Scholes exactly; the put comes from parity, so no second formula is needed”

C2: Critique an AI Answer, Task [Proposed]

Solution discussed in class

Task

- The answer contains **three** errors; everything else is correct
- For each error
 - ▶ say what is wrong and why
 - ▶ say how you would detect it: a no-arbitrage bound, a unit check, the lecture, a second method
 - ▶ give the corrected line
- Run the corrected code; report C and P and check put–call parity numerically
- Log the three errors as in `AI_ERRORS.md`: what was asked, what the AI answered, what was wrong, how you found out, the correct version

Interpretation

- Before running the code: what should a 90-day call 5% out of the money cost, roughly: 0.2, 2 or 20?
- Which error would still give a plausible-looking price? Which check catches it?
- Why is “the put comes from parity” not a check of the call price?

C3: Testing the Martin Bound (1/2) [Proposed]

Solution discussed in class

□ Context and data

- ▶ Lecture 12, case study: [Back, Crotty & Kazempour \(2022\)](#) test the bound of [Martin \(2017\)](#), $E_t[R_{t,T}^e] \geq b_{t,T}$, with the Kodde–Palm statistics D_1 (validity) and D_2 (tightness) [Kodde & Palm \(1986\)](#); $R_{t,T}^e$ is the market excess return from the close of day t over h months
- ▶ Data: the authors' daily Martin bound, annualised %, $h = 1, 3, 6, 12$, 2 January 1990 – 31 December 2020 (columns 1b_m_h of `market_bounds.csv`, [replication package](#)); S&P 500 daily closes (GSPC.INDX in `data/market`); the daily risk-free rate ([Kenneth French Data Library](#)); the monthly Welch–Goyal predictors [Welch & Goyal \(2008\)](#) ([Amit Goyal's data page](#))

□ Tasks

- ▶ (a) Compute the daily excess return of the S&P 500 (price return minus the risk-free rate), compound it over the next 21, 63, 126 and 252 trading days (the empirical convention of the authors, close to but not equal to $R_{t,T} - R_{f,t,T}$), annualise it with the factor $12/h$, in %; use returns up to March 2021 and formation dates 31 January 1990 – December 2020 with the bound, the complete forward return and the predictors of (b) available (Table 1 sample, $N = 7789$ at 1 month, 7598 at 12 months); compare the means with Table 1: the price index has no dividends, so the return means will be lower
- ▶ (b) Build the nine positive conditioning variables as defined in the notes to Table 3 (log dividend-price and earnings-price ratios plus 5, book-to-market, T-bill rate, $1 +$ term spread, credit spread, stock variance, $1 +$ net equity issuance, $1 +$ inflation), assign each month to its last trading day and carry it forward; form the 10 moments $s_{t,T}(1, z_t)$ with $s_{t,T} = R_{t,T}^e - b_{t,T}$ and their sample mean $\bar{\lambda}$

C3: Testing the Martin Bound (2/2) [Proposed]

Solution discussed in class

▣ Tasks

- ▶ (c) Estimate Σ , the covariance of $\bar{\lambda}$, by Hansen–Hodrick [Hansen & Hodrick \(1980\)](#) with $21h$ lags (Newey–West with $1.5 \times 21h$ lags if it is not positive semi-definite); compute D_1 and D_2 with a nonnegative least-squares solver and their asymptotic p -values from the χ^2 mixture, with weights simulated as in Appendix A
- ▶ (d) Regress $R_{t,T}^e$ on a constant and $b_{t,T}$ with Hansen–Hodrick errors ($21h$ lags) and test $H_0: \alpha = 0, \beta = 1$ jointly; then repeat (c) with the bound replaced by its time-series mean, as in Section 4.3

▣ Report and interpret

- ▶ A table with your means, D_1 , D_2 , p -values and the joint-test p -values next to Tables 1 and 3 of the paper; explain every discrepancy that cannot be attributed to rounding
- ▶ Interpretation: why can “ $\alpha = 0, \beta = 1$ is not rejected” coexist with “tightness is rejected”? Does validity survive the mean-bound test of (d), and what does that say about the evidence for the bound?
- ▶ Model: B5 (regression with overlapping windows and HAC errors)

Common Pitfalls

- ▣ Forgetting dividends or the interest rate in parity: a false arbitrage
- ▣ Treating the risk-neutral probability as a forecast
- ▣ Quoting vega or theta without the unit (per vol point? per day? per year?)
- ▣ Newton's method on deep OTM options ($\text{vega} \approx 0$): use a bracketing solver
- ▣ Naive standard errors on overlapping windows: t -statistics several times too large
- ▣ Reading a high Sharpe ratio of a short-volatility strategy without its skewness and worst month

Take-Home Assignment (Team Project)

- ▣ Take a new Deribit snapshot for Ether (ETH) options and build the smile of three expiries with SVI, with bootstrap intervals
- ▣ Compute a 30-day VIX-type index and the risk-neutral density; compare with Bitcoin [Pele et al. \(2023\)](#)
- ▣ Delta-hedge a sold one-week ATM Bitcoin option on historical paths since March 2021 [Matic, Packham & Härdle \(2023\)](#)
 - ▶ Price and hedge each option at the DVOL of its trade date (the information then available), $r = 0$, daily rebalancing at the 00:00 UTC closes, non-overlapping weekly windows
 - ▶ Report the mean and standard deviation of the error in % of the premium, with moving-block bootstrap intervals; justify every choice you change
- ▣ Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations
- ▣ Log every AI error you catch in `AI_ERRORS.md`, as in C2: what was asked, what was wrong, how you found out, the correct version

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  XFG Advanced Methods in Quantitative Finance
- ▣  Pricing Kernels and Risk Premia
- ▣  Hedging Cryptocurrency Options

Code and data

- ▣ Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_12

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